**Software Requirement Specification (SRS)**

For

# Project: AI Assisted Smart Library System

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# 1.Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to document the functional and non- functional requirements for the "NextGen Smart Library System." This document is intended for developers, project stakeholders (guides/evaluators), and future maintainers of the system. It outlines the architectural decisions, specifically focusing on the integration of Artificial Intelligence for resource discovery and temporal logic for resource scheduling.

## 1.2 Intended Customers/Users

The system is designed for the following user classes within an academic institution:

- **Students:** The primary beneficiaries who will use the system to discover study materials and book physical resources.
- **Faculty or Research Scholars:** Users requiring access to niche academic resources.
- **Librarians as Administrators:** Personnel responsible for managing inventory, approving requests, and maintaining system integrity.

## 1.3 Product Scope

The "AI Assisted Smart Library System" is a full-stack web application designed to modernize traditional library operations. Unlike standard CRUD (Create, Read, Update, Delete) systems, this product integrates:

- **AI-Driven Discoverability:** A search engine capable of understanding user intent despite spelling errors (Fuzzy Logic) and recommending books based on content similarity (TF-IDF).
- **Resource Planning:** A rule-based scheduling engine for discussion rooms that prevents conflicts and enforces institutional policies (e.g., ID validation).
- **Administrative Resilience:** A disaster recovery module ("Black Box") to prevent accidental data loss.

## 1.4 Outline of the Document

This document is organized into specific sections: Section 2 describes the general factors that affect the product and its requirements. Section 3 (omitted as per instruction) covers system models. Section 4 details interface requirements. Section 5 specifies non-functional requirements like security and performance. Section 6 outlines the testing strategies used to validate the software.

# 2.Overall Description

## 2.1 Product Perspective

This system is a standalone web-based solution. It is built using the Flask micro-framework (Python) and acts as a replacement for manual ledger systems or legacy library software that lacks "intelligence." It utilizes a flat-file database architecture (CSV) for portability during the prototype phase but is architected to support migration to SQL databases.

## 2.2 Product Functions

The major functions of the system include:

1. Smart Search: Retrieving book results using Natural Language Processing (NLP) to handle typos and context.
2. Recommendation Engine: Suggesting relevant reading materials based on the category and content of the currently viewed book.
3. Temporal Room Booking: Allowing student groups to reserve discussion rooms with strict validation (Time, ID, Concurrency).
4. Inventory Management: Adding, updating, or removing books from the catalog.
5. Disaster Recovery: Automated backup of critical data before bulk administrative actions.

## 2.3 User Classes and Characteristics

**Student User:**

- Technical Skill: Basic web navigation.
- Access Rights: Search, View Recommendations, Book Rooms.
- Constraint: Must possess a valid Registration Number in the format 24bca7976.

**Administrator:**

- Technical Skill: Moderate. Capable of understanding system alerts.
- Access Rights: Full CRUD access to Inventory and Bookings. Capability to override or cancel active bookings.

## 2.4 Design and Implementation Constraints

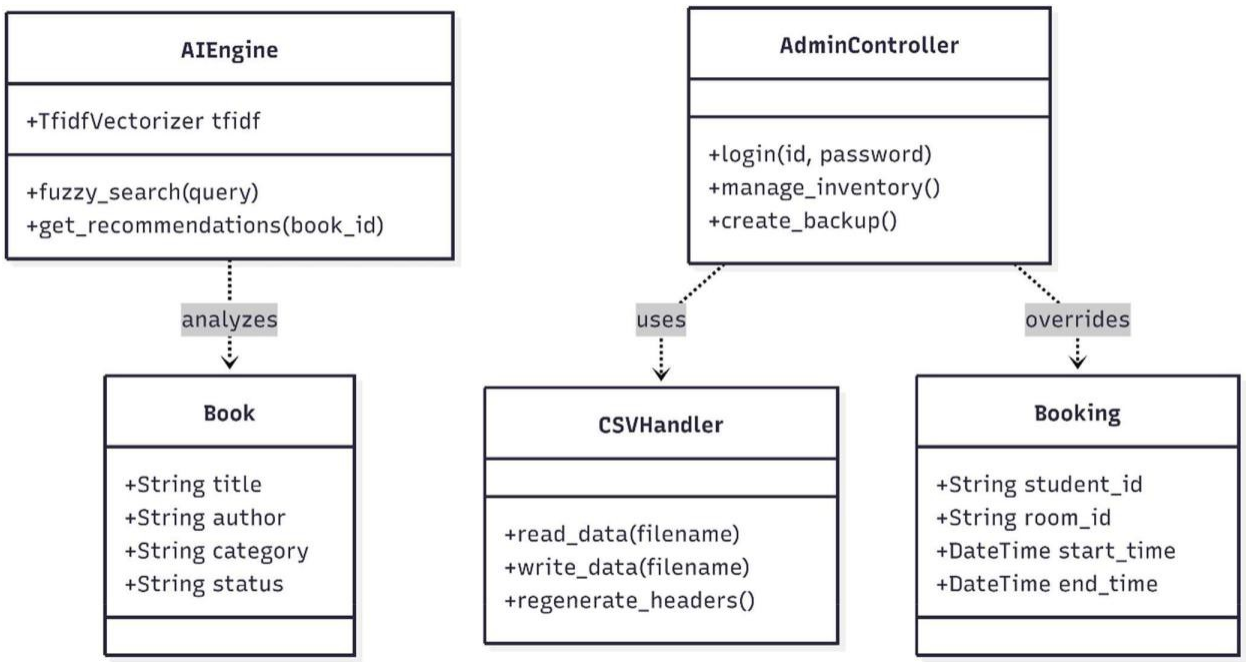
- Hardware: The system must run on standard university lab computers with minimal RAM overhead (AI models are pre-computed).
- Network: Requires an Intranet or Internet connection for the Web UI.
- Persistence: For this version (V1.0), data is stored in CSV format. Concurrent write operations are limited by file-locking mechanisms.
- Time Constraints: Room bookings are strictly limited to the operational hours of 09:00 AM to 18:00 PM.

## 2.5 Assumptions and Dependencies

- It is assumed the hosting server has Python 3.8+ installed.
- The system relies on open-source libraries: scikit-learn, pandas, fuzzywuzzy. It is assumed these libraries remain compatible with the deployment environment.
- User ID format policies (Batch+Branch+Serial) are assumed to remain constant.

## 3.Diagrams (UML, ER, DFD)

### 1.CLASS_DIAGRAM:



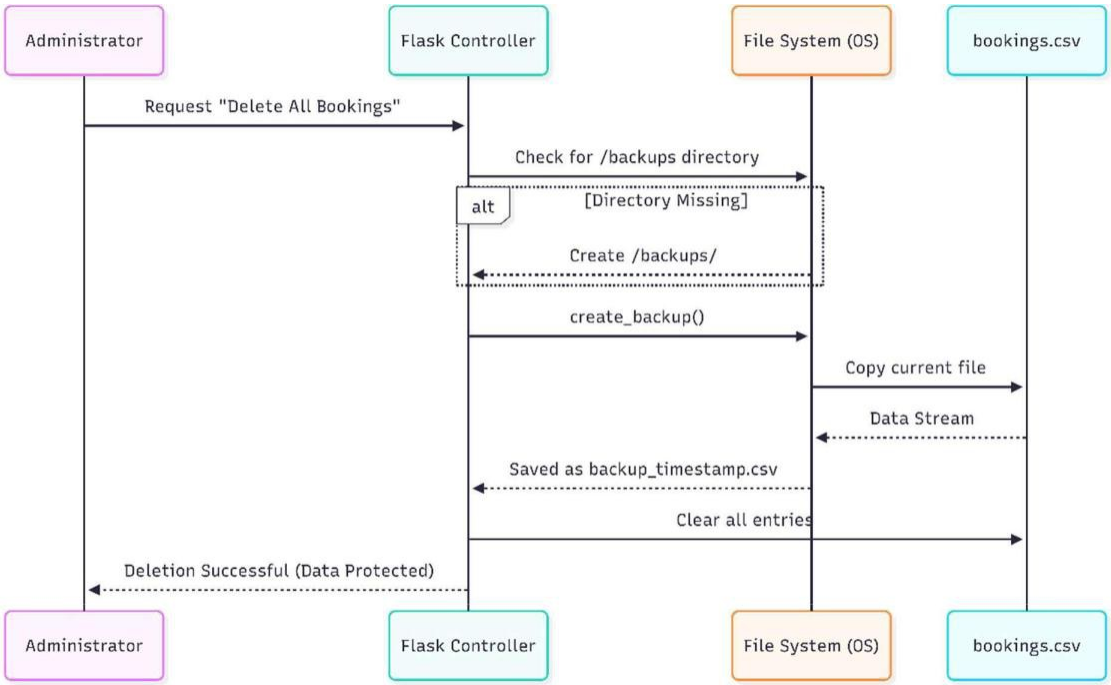

**Description:**

This is a static model that illustrates the internal structure of the software, including its attributes and methods.

**Purpose:**

It shows how the **AI Engine** interacts with the **Book** data and how the **Admin Controller** utilizes the **CSV Handler** to manage data persistence and backups.

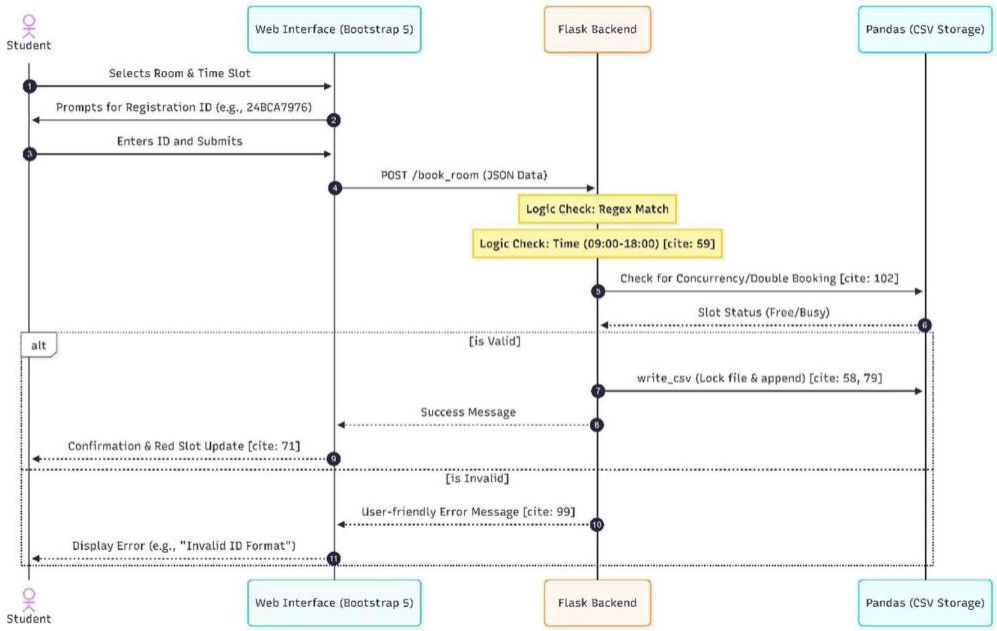
## 2.SEQUENCE_DIAGRAM FOR _RECOVERY:




## <u>Disaster Recovery:</u>

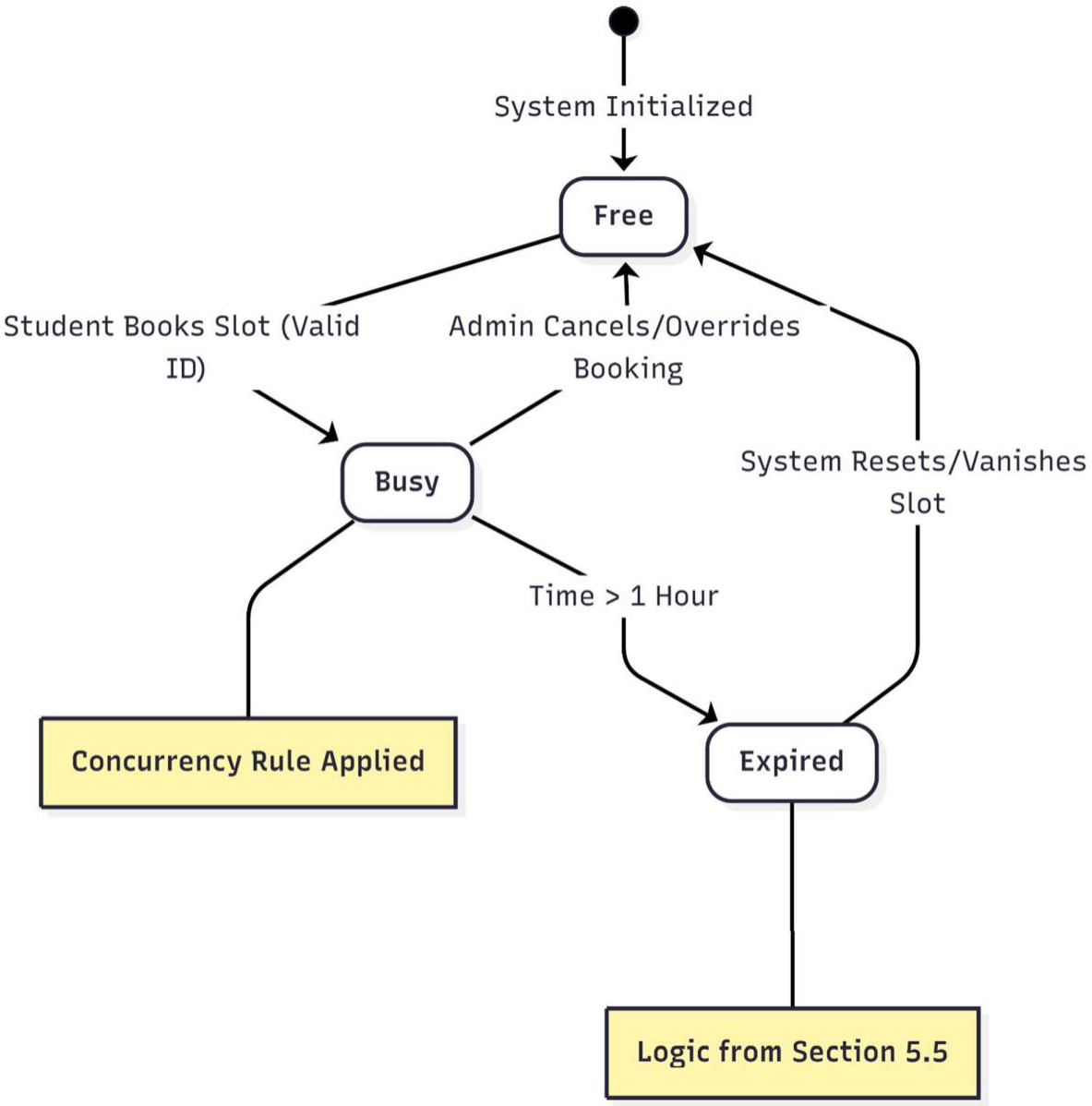
This focuses on the **"Black Box"** safety mechanism, showing how the system creates an automated data backup before executing bulk administrative deletion

## 3 SEQUENCE_DIAGRAM for Discussion Room:




**<u>Discussion Room Booking:</u>** This diagram illustrates the chronological flow of data between the User Interface, the Flask Backend, and the CSV database. It ensures that the **1-hour booking rule** and **operational hour constraints (09:00 - 18:00)** are enforced.

## 4.STATE_MACHINE Diagram (Booking):



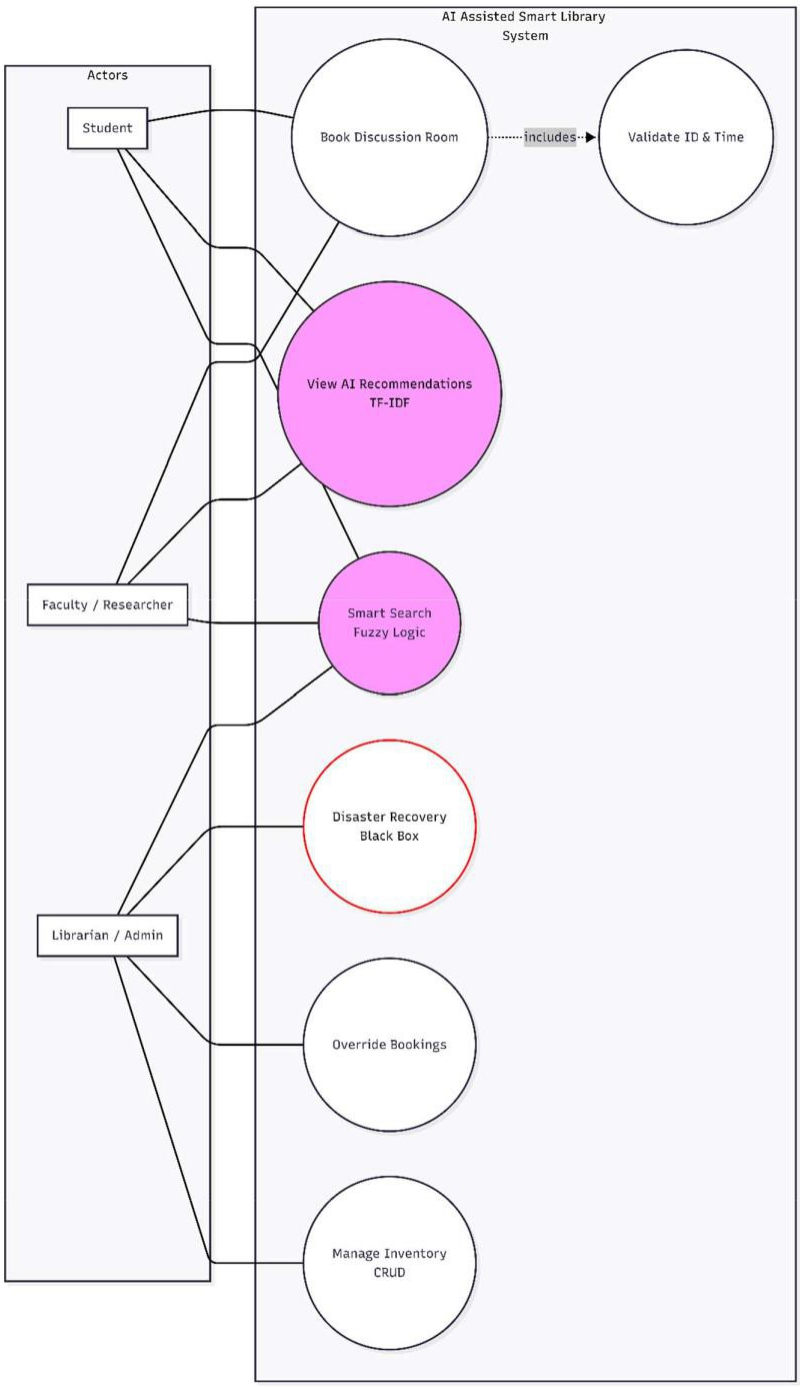

**Description:**

This diagram tracks the different states of a room booking slot during its lifecycle.

**Purpose:**

It manages the transition from `Free` to `Busy` and ensures that slots move to an `Expired` state once the **one- hour time limit** is reached.

## 5.USE_CASE Diagram:

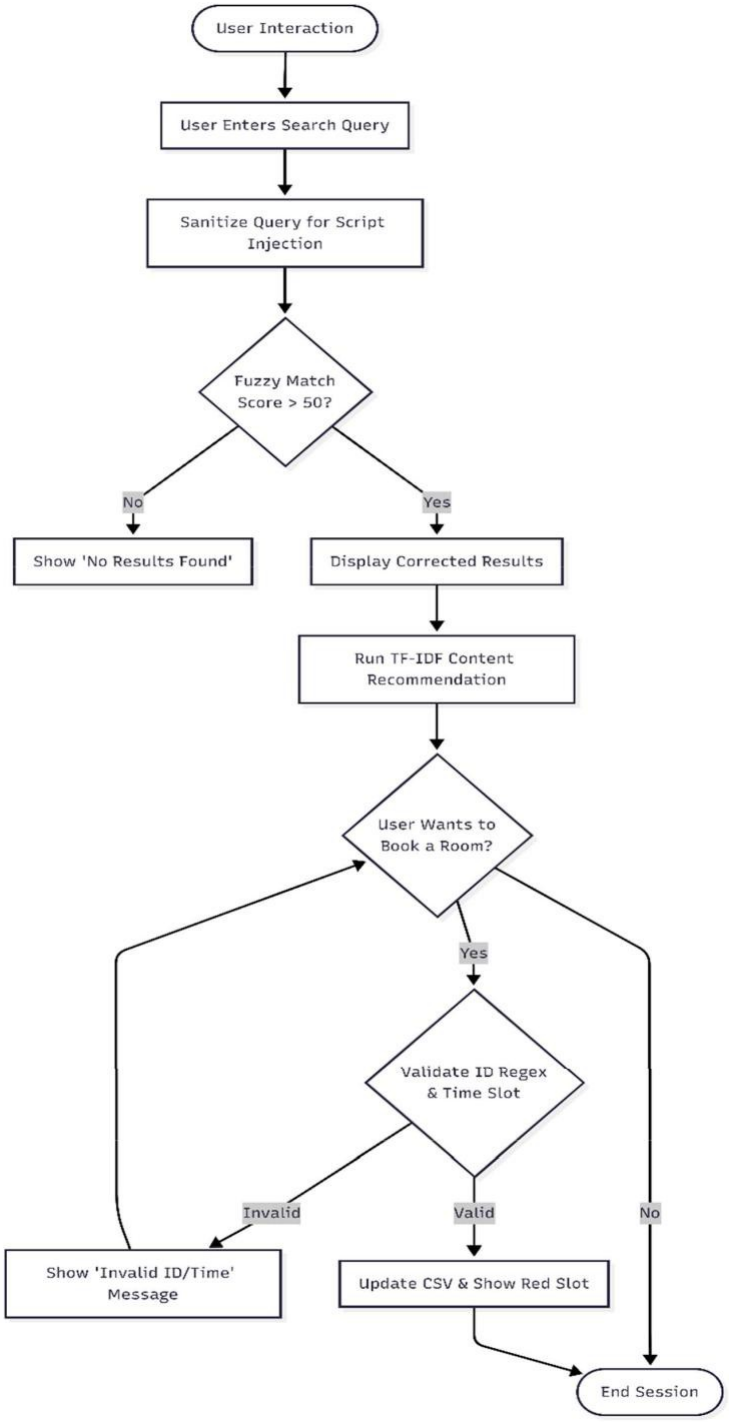



**Description:**

This diagram defines the functional scope of the system by illustrating how different actors—Students, Faculty, and Administrators—interact with specific features.

**Purpose:** It highlights the unique AI-driven services, such as **Fuzzy Logic search** and **TF-IDF recommendations**, alongside critical administrative functions like **Disaster Recovery**

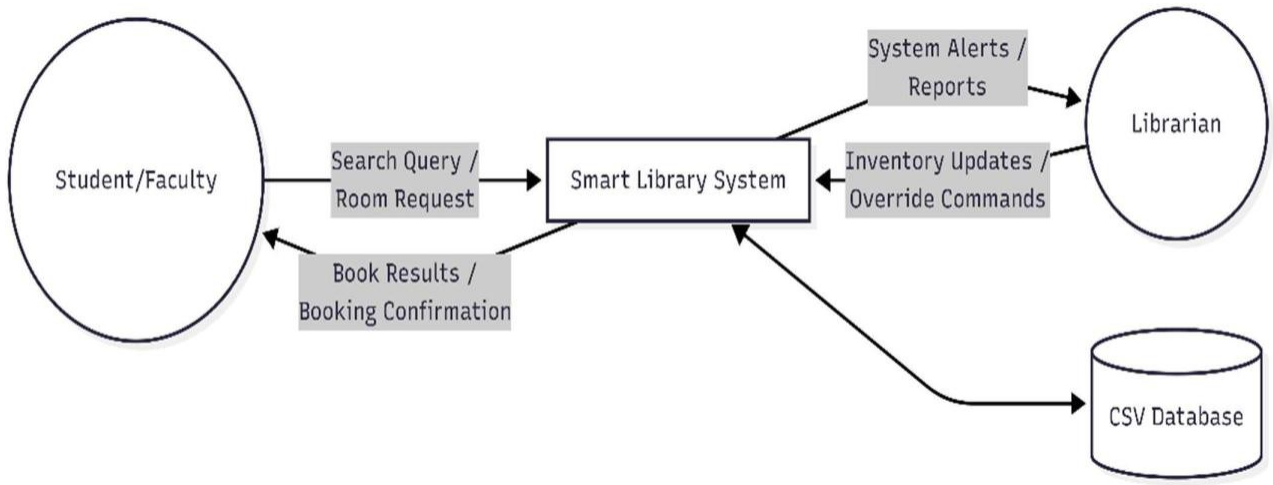
## 6. Activity Diagram for AI search:




**<u>Description:</u>** This dynamic model tracks the step-by-step logic of the system during a search and booking session.

**<u>Purpose:</u>** It specifically models the **Fuzzy Match threshold logic (Score > 50)** and the **Regex validation**
path required to finalize a room reservation

## 7.Data Flow Diagram:



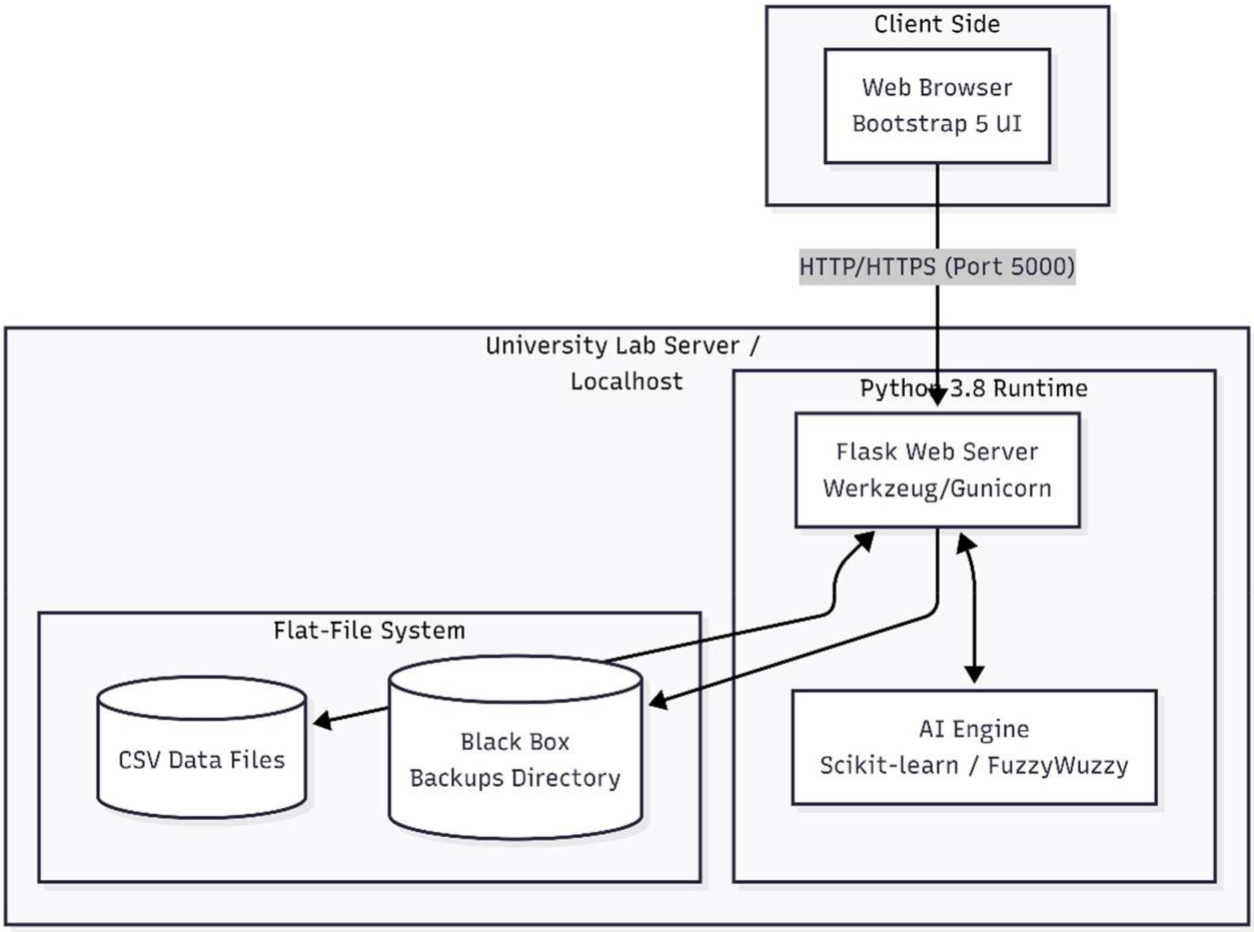

**Description:**

This diagram provides a high-level view of the entire data ecosystem.

**Purpose:**

It identifies the data flow between external entities (Students and Librarians) and the **Smart Library System**, showing exactly what information is input and output

## 8.Deployment Diagram:



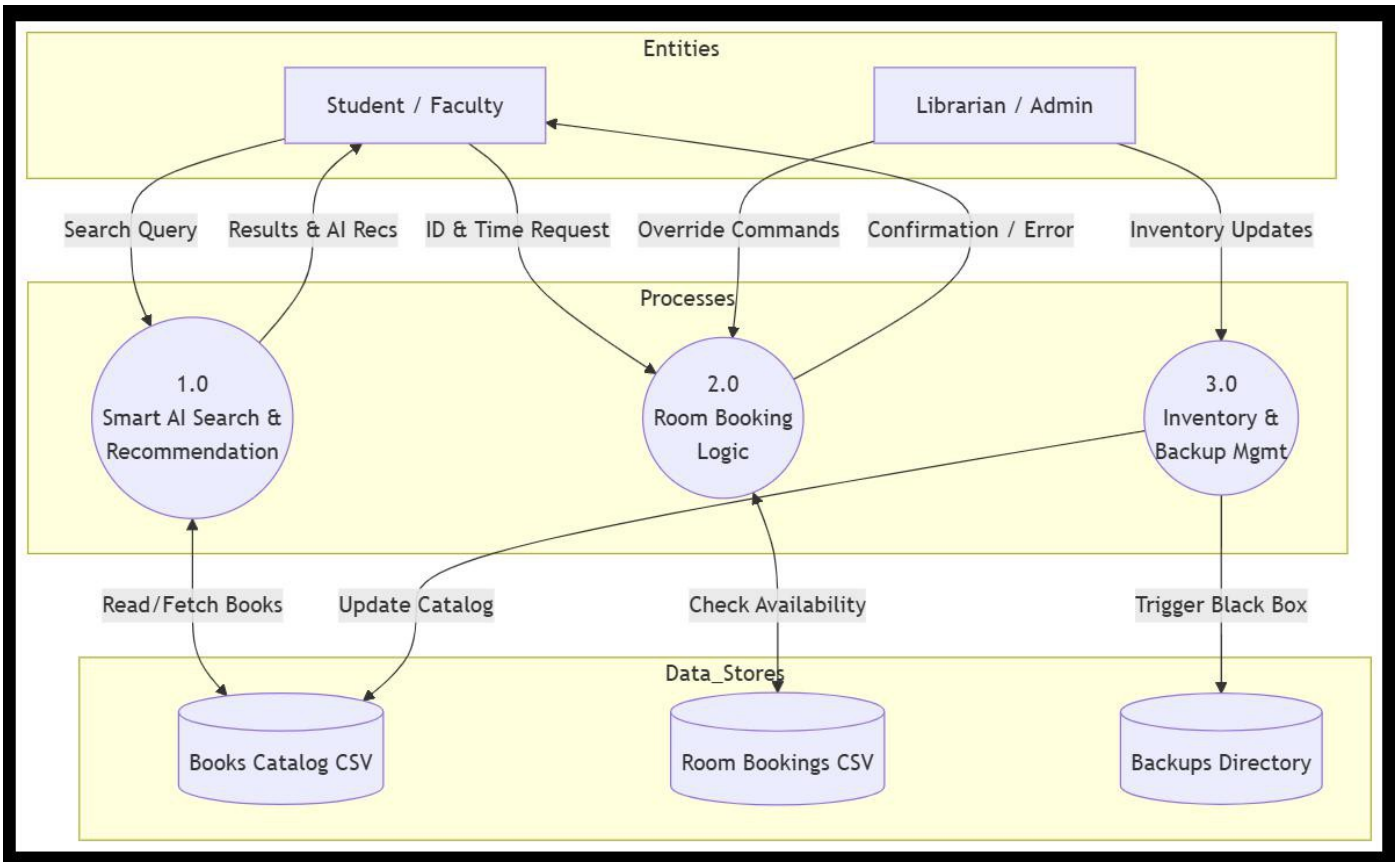

**Description:**

This model shows the physical hardware and software environment where the application is hosted.

**Purpose:**

It visualizes the connection between the **Bootstrap 5 client** and the **Python/Flask server**, detailing how they interact with the **Flat-File (CSV) storage system**

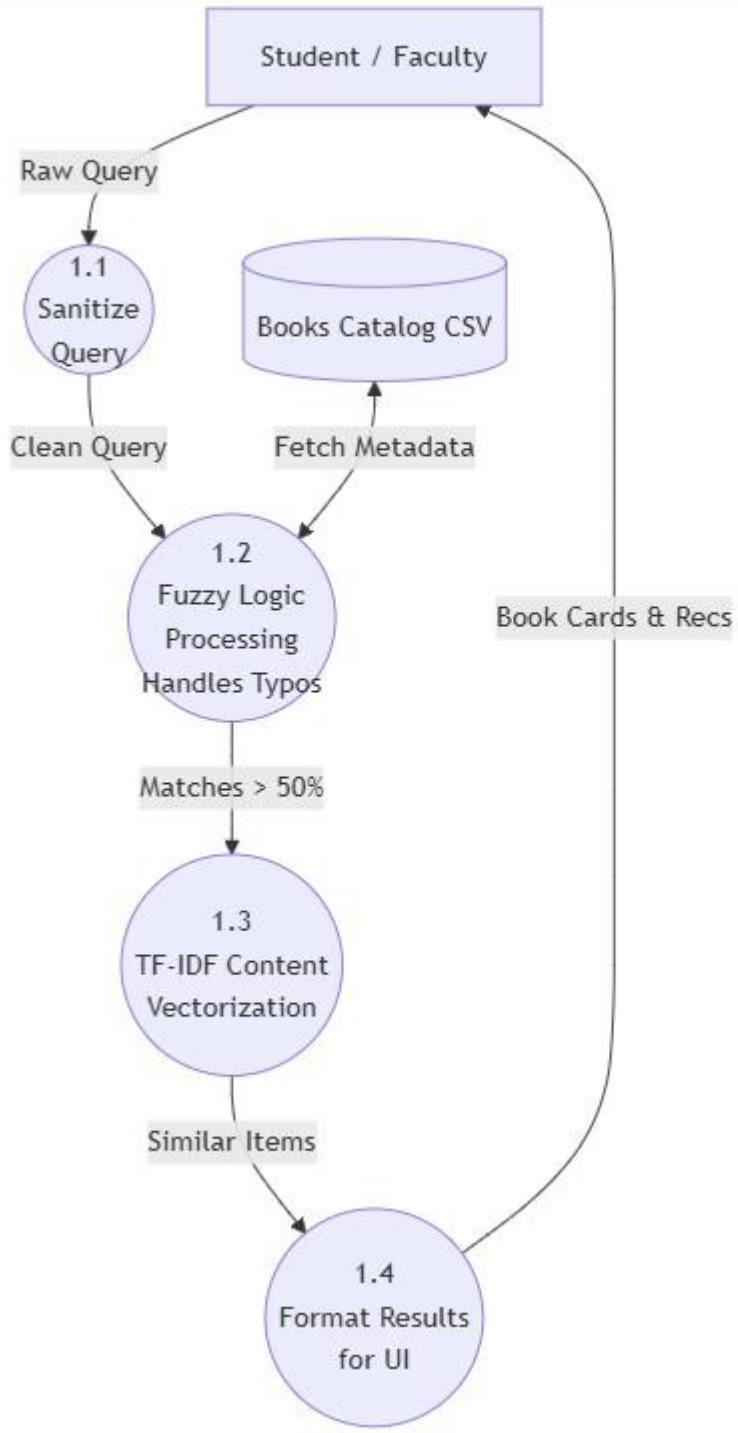
## 9.Data flow Diagram -level 1




**DFD Level 1 (Major Subsystems):**

Decomposes the central system into three distinct modules: AI-Driven Search, Resource Scheduling, and Administrative Management. It illustrates how the Student and Librarian actors interact with specific data stores like the Books Catalog and Room Bookings CSV files.

## 10.Data flow Diagram -level 2




**DFD Level 2 (AI Process Decomposition):**

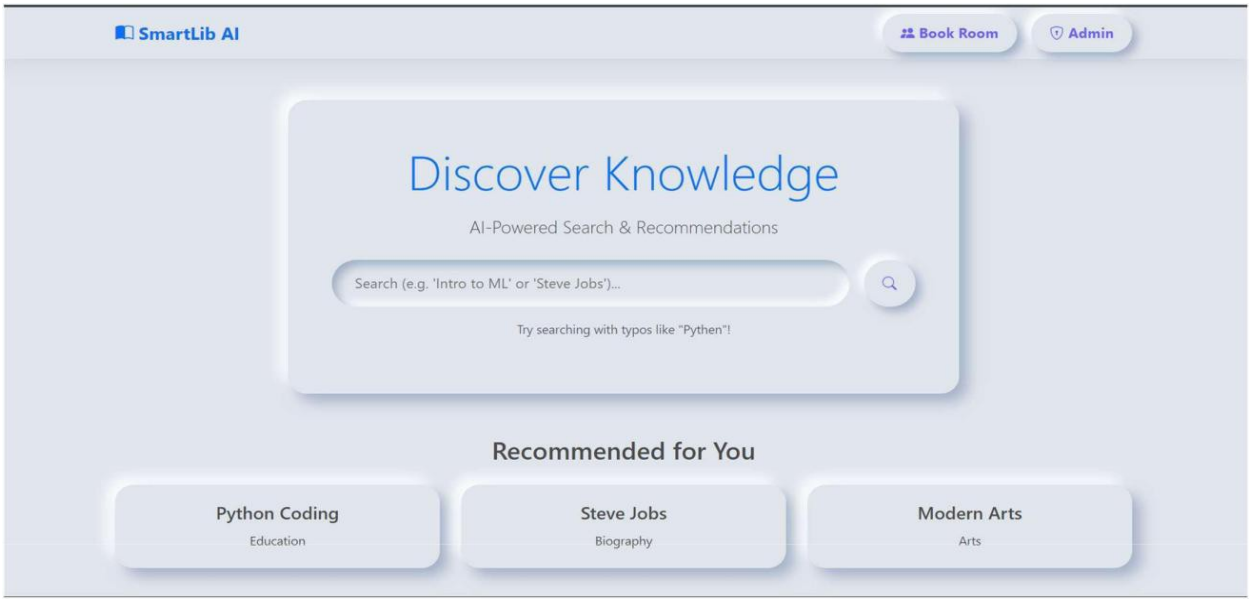
Provides a granular view of the **Search Engine (Process 1.0)**. It highlights the internal data transformations—from initial query sanitization to **Fuzzy Logic** matching and **TF-IDF** recommendation generation—that ensure high resource discoverability.

# 4. External Interface Requirements

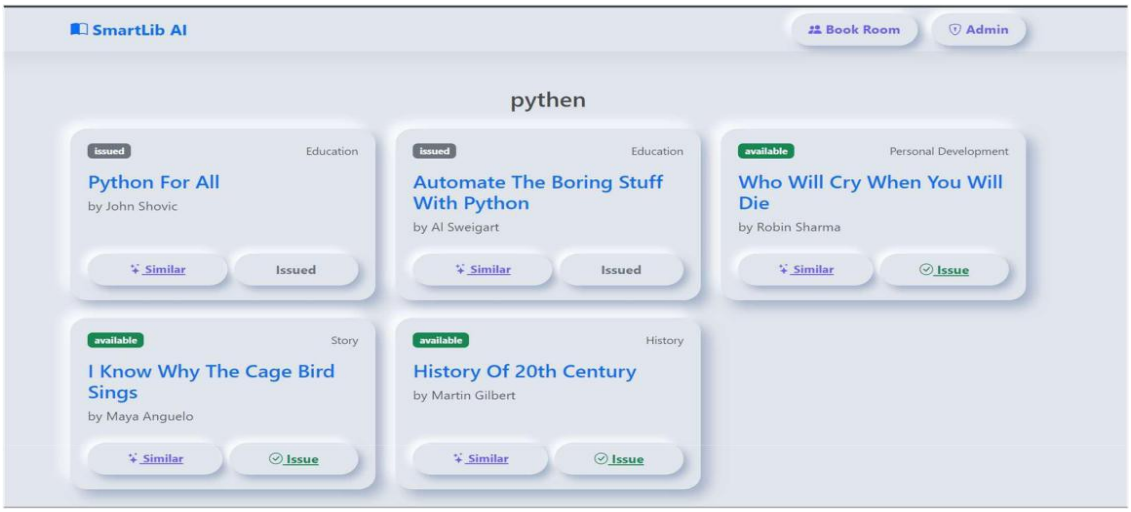
## 4.1 User Interfaces

The system features a Neumorphic (Soft UI) design language implemented via Bootstrap 5 and custom CSS.

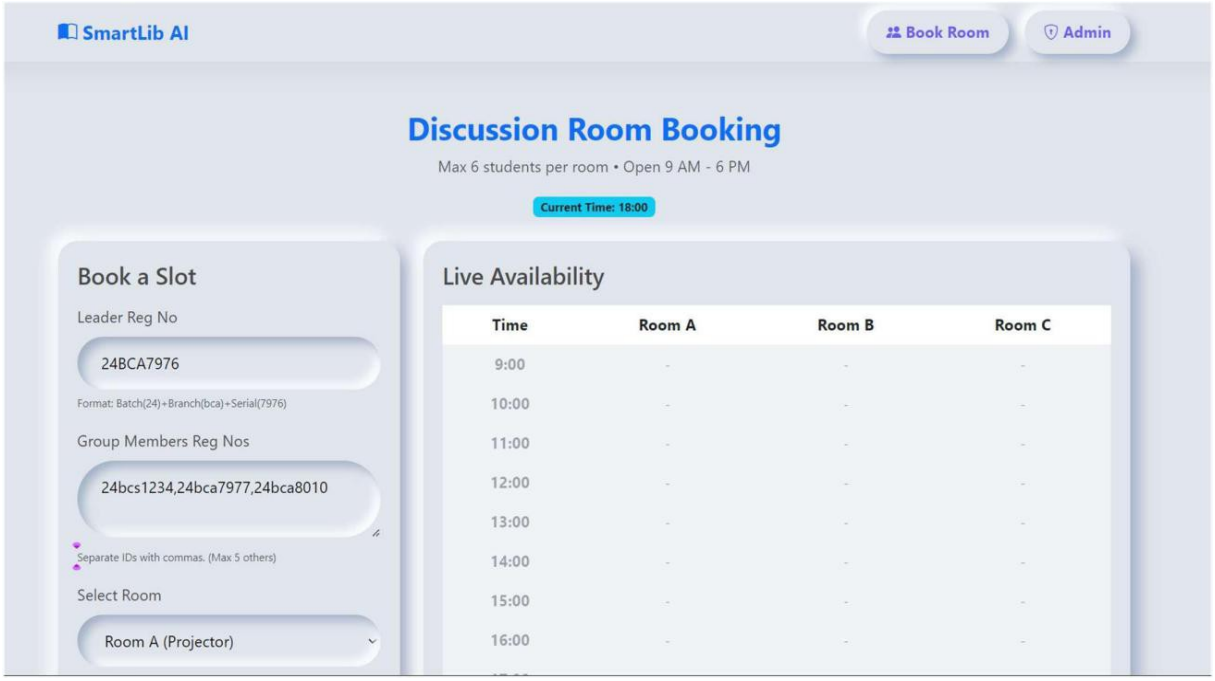
- Home Screen: Features a prominent "Smart Search" bar and "Quick Actions" for room booking.



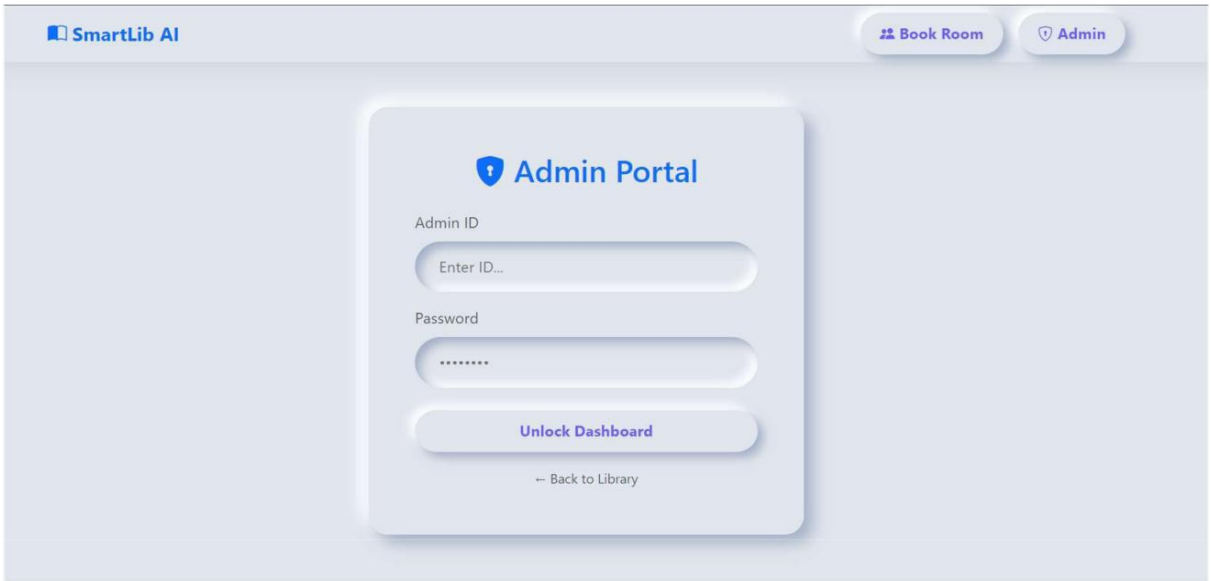

- Search Results Page: Displays books in a card layout. Each card contains "Title", "Author", "Status Badge", and a "Recommend Similar" button.



- Room Booking Grid: A responsive table showing time slots (Rows) vs. Rooms (Columns). "Busy" slots are visually distinct (Red) from "Free" slots (Green).




- Admin Dashboard: A dense data table view for inventory management with inline "Quick Action" buttons for status updates.



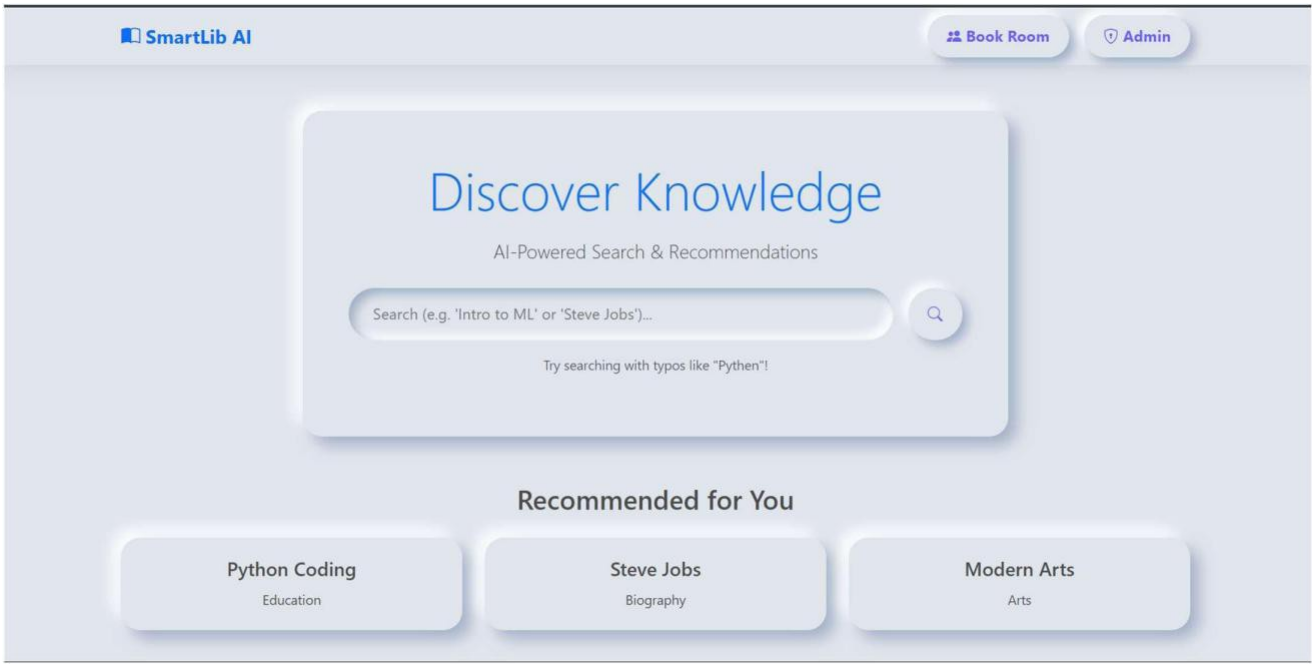
## 4.2 Hardware Interfaces

- Server Side: No specific hardware dependency; runs on standard x86/x64 architecture.
- Client Side: Responsive design supports Desktops (1920x1080), Laptops (1366x768), and Tablets

DESKTOP

## 4.3 Software Interfaces

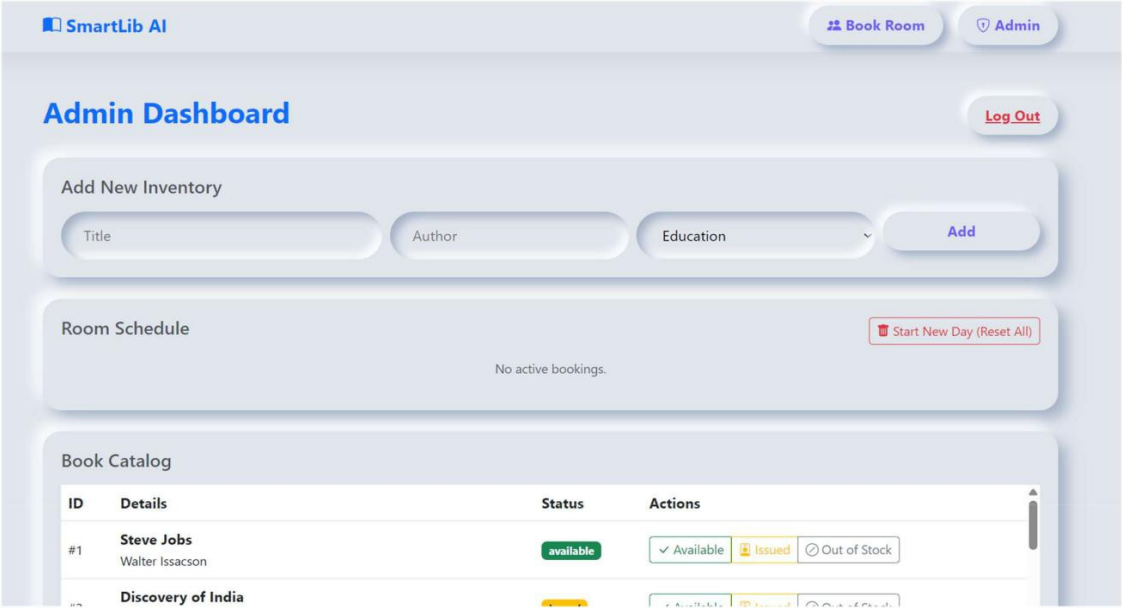
- Operating System: Cross-platform compatibility (Windows/Linux/macOS).
- Web Server: Werkzeug for Development ,and Gunicorn for Production.



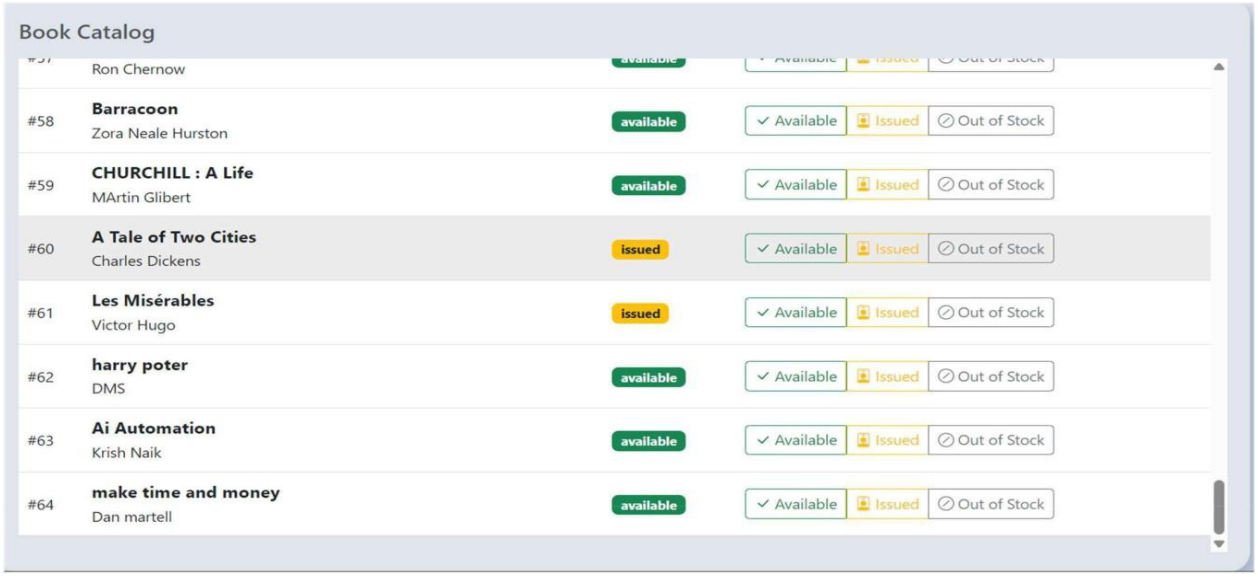

- Data Processing: Pandas DataFrames acting as an in-memory database interface.

## 4.4 Communication Interfaces

- Protocol: HTTP/HTTPS (Port 5000 default).
- Data Format: JSON is used for asynchronous (AJAX) communication between the Admin Frontend and the Flask Backend to update book statuses without page reloads.

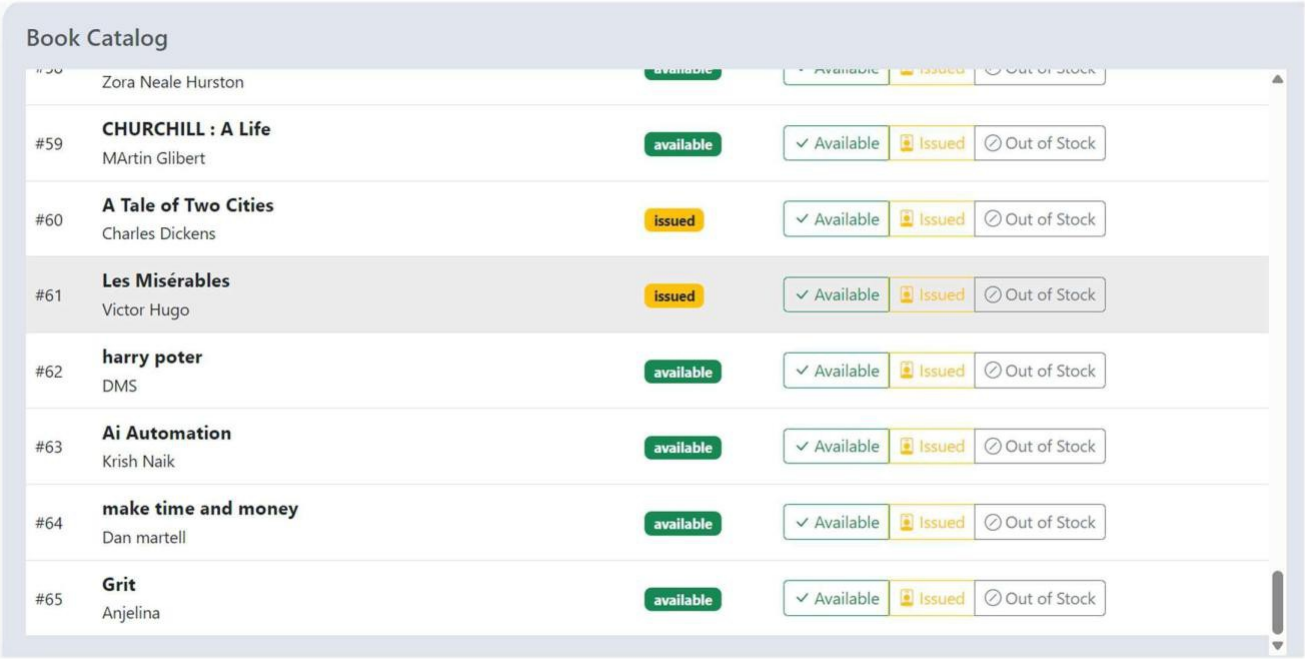



**Book Catalog:**



## Admin Dashboard

## Admin added new Book: Grit successfully



# 5.Other Non-Functional Requirements

## 5.1 Performance Requirements

- Search Latency: The AI search query must return results in under 2.0 seconds for a dataset of 5,000 books.
- Startup Time: The TF-IDF Vectorization model must train and initialize within 15 seconds of server boot.

## 5.2 Safety Requirements

- Data Preservation: The system must automatically create a timestamped copy of the bookings.csv file in the backups directory before any "Delete All" operation is executed.
- Fail-Safe: If the CSV file is corrupted or missing, the system must regenerate headers automatically to prevent crash loops.

## 5.3 Security Requirements

- Authentication: Administrative access is restricted via a session-based login requiring a **username (24bca7976) and password.**
- Input Validation:

  Student IDs are validated against the

  **Regex pattern: ^\d{2}[a-zA-Z]{3}\d{4}$.**

  Search queries are sanitized to prevent potential script

  injection.

- Access Control: Direct URL access to /admin routes without an active session is blocked and redirected to the login page.

### 5.4 Software Quality Attributes

- Robustness: The system handles edge cases
- e.g., booking a time slot that has already passed

with user-friendly error messages rather than stack traces.

- Usability: The "Fuzzy Search" feature ensures that users with low spelling proficiency can still utilize the library effectively.

### 5.5 Business Policies & Rules

- Concurrency Rule: A student cannot be part of two room bookings at the same time.
- Capacity Rule: A single discussion room cannot accommodate more than 6 students.
- Expiry Rule: Booking slots are valid for 1 hour; expired slots must visually "vanish" or be disabled in the UI.

# 6.Testing

## 6.1 Black-Box Testing

Focuses on input/output without looking at code.

| Test ID | Test Case Description | Input Data | Expected Result |
|---|---|---|---|
| BB-01 | Typo Tolerance Search | Query: "Pythen" | System displays "Python" related books. |
| BB-02 | Past Time Booking | Time: 10:00 AM (Current Time: 2:00 PM) | Error: "Cannot book past time slots." |
| BB-03 | Invalid ID Format | ID: "User123" | Error: "Invalid ID Format. Must match 24bca7976." |
| BB-04 | Double Booking | Student A books Room 1, then tries Room 2 | Error: "Student is already part of another booking." |
| BB-05 | Admin Login | User: 24bca7976, Pass: Wrong | Access Denied. Flash message displayed. |

### 6.2 White-Box Testing

Focuses on internal logic and code structure.

- WB-01 (Regex Logic): Verified the re.match() function correctly rejects strings with special characters or incorrect lengths in validate_booking().
- WB-02 (Vectorization): Verified that TfidfVectorizer correctly ignores English stop-words (e.g., "the", "and") during the matrix generation in ai_engine.py.
- WB-03 (File I/O): Verified that the create_backup() function correctly checks for the existence of the backups/ directory and creates it if missing before copying files.
- WB-04 (Levenshtein Distance): Tested the process.extract() threshold; confirmed that matches with a score < 50 are discarded to maintain relevant.

## 7. Conclusion

The "AI Assisted Smart Library System" successfully bridges the gap between traditional static library catalogues and modern, user-centric expectations. By integrating AI for "forgiving" search and contextual recommendations, the system significantly improves resource discoverability. Furthermore, the logic-driven booking system eliminates administrative overhead by enforcing rules automatically. The prototype demonstrates that intelligent features can be delivered in a lightweight, portable architecture, setting the stage for future scalability.

## 8.References


- Flask Documentation (v2.0): https://flask.palletsprojects.com/
- Scikit-Learn User Guide (TF-IDF): https://scikit-learn.org/
- FuzzyWuzzy Library: https://github.com/seatgeek/fuzzywuzzy
- Bootstrap 5 Documentation: https://getbootstrap.com/


## 9.Acknowledgements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